Dmytro Kuzmenko

PhD Candidate in Computer Science | Assistant Professor, NaUKMA

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Dmytro Kuzmenko is a machine learning researcher working on embodied intelligence that is efficient enough to run under real compute budgets and trustworthy enough to act around people. His work spans world models and model-based RL, modular instruction-conditioned policies for efficient embodied AI, and safety-aware robot behaviour. Earlier work was on semi-supervised perception for assistive and wearable locomotion systems. He expects to defend his PhD in September 2026 and is a Visiting PhD Researcher at AIRlab, University of Turin.

Research Interests

World models & model-based RL: latent-dynamics world models, learning coupled with model-predictive control, policy distillation and compute-constrained inference

Foundation models for autonomy: VLM/VLA systems for embodied reasoning, instruction routing and grounding, spatial and deictic reference resolution

Physical AI safety: safety- and ethics-aware refusal and override behaviour, proxemic risk assessment from egocentric robot perception, HRI evaluation protocols

Reinforcement and imitation learning: multi-task RL, knowledge transfer between agents, reward shaping under physical constraints

Distribution shift & data efficiency: sim-to-real digital-twin calibration under query budgets, semi-supervised perception, zero-shot transfer in navigation

Multi-agent coordination: cooperation under partial observability in open, ad-hoc teams

Education

PhD in Computer Science *2023 – Sep 2026 (expected)*
National University of Kyiv-Mohyla Academy (NaUKMA), Kyiv, Ukraine.
Advisor: Dr. Nadiya Shvai. Expected defence: September 2026.
Dissertation: *Resource-Aware Embodied Intelligence Through the Optimisation of Data, Compute, and Trust.*

MSc in Applied Mathematics *2021 – 2023*
National University of Kyiv-Mohyla Academy (NaUKMA), Kyiv, Ukraine.

BSc in Software Engineering *2016 – 2021*
National University of Kyiv-Mohyla Academy (NaUKMA), Kyiv, Ukraine.

Academic & Research Appointments

AIRlab, Department of Computer Science, University of Turin Turin, Italy
Visiting PhD Researcher Feb 2026 – present
PI: Prof. Cristina Gena. Research on assistive and human-aware robotics, human-robot teaming, and HRI evaluation. Contributor to *This/That: Spatial Reference Resolution for VLMs* (ongoing); advisor on a completed 3D scene-mapping project for a museum tour-guide robot. Deployed and integrated Reachy Mini as a shared experimental platform for the lab.

Department of Multimedia Systems, NaUKMA Kyiv, Ukraine (Remote)
Assistant Professor Oct 2022 – present
Designed and delivered seven graduate and undergraduate courses in machine learning, computer vision, reinforcement learning, and NLP. Built and run an independent student research program to develop research directions, convert them into tractable projects, and mentor students through experimentation, writing, rebuttals, and publication; one supervised MSc thesis, 12 supervised BSc theses, 10 supervised course papers, and the first two student-led papers accepted at ECCV 2026 workshops.

Machine Intelligence Lab, University of Toronto Toronto, Canada (Remote)
Research Assistant Apr 2022 – Oct 2023
Supervised by Dr. Brokoslaw Laschowski. Efficient visual perception for robotic exoskeletons and wearable locomotion systems; semi-supervised stair-recognition pipelines for human-robot walking environments. Published at IROS 2023 and BioMedical Engineering OnLine 2024.

Publications

Peer-Reviewed Journal Articles

1. Kuzmenko, D., Shvai, N. "MoIRA: Modular Instruction Routing Architecture for Multi-Task Robotics." *Neurocomputing*, vol. 674, 132962, 2026.
   doi:10.1016/j.neucom.2026.132962 | arXiv:2507.01843
2. Kurbis, A. G., Kuzmenko, D., Ivanyuk-Skulskiy, B., Mihailidis, A., Laschowski, B. "StairNet: Visual Recognition of Stairs for Human-Robot Locomotion." *BioMedical Engineering OnLine*, 23(1):20, 2024.
   doi:10.1186/s12938-024-01216-0
3. Beimuk, V., Kuzmenko, D. "Energy Conservation for Autonomous Agents Using Reinforcement Learning." *NaUKMA Research Papers: Computer Science*, vol. 8, pp. 68–75, 2025.
   doi:10.18523/2617-3808.2025.8.68-75

Peer-Reviewed Conference Papers

1. Kuzmenko, D., Shvai, N. "When Robots Say No: The Empathic Ethical Disobedience Benchmark." In *Proc. ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Edinburgh, Scotland, pp. 168–176, March 2026. *Main track.*
   doi:10.1145/3757279.3785547 | arXiv:2512.18474 | code | project page
2. Kuzmenko, D., Shvai, N. "Knowledge Transfer in Model-Based RL Agents for Efficient Multi-Task Learning." In *Proc. 24th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, Detroit, MI, USA, pp. 2597–2599, May 2025.
   doi:10.5555/3709347.3743949 | arXiv:2501.05329
3. Kuzmenko, D., Shvai, N. "Balancing Performance and Efficiency in Zero-shot Robotic Navigation." In *Proc. 19th International Conference on ICT in Education, Research and Industrial Applications (ICTERI 2024)*, Springer CCIS, 2025.
   doi:10.1007/978-3-031-81372-6_28 | arXiv:2406.03015
4. Kuzmenko, D., Tsepa, O., Kurbis, A. G., Mihailidis, A., Laschowski, B. "Efficient Visual Perception of Human-Robot Walking Environments using Semi-Supervised Learning." In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, MI, USA, October 2023.
   doi:10.1109/IROS55552.2023.10341654

Student-Led Workshop Papers — supervising author

1. Rudas, V., Kuzmenko, D. "Can Vision-Language Models Assess Proxemic Risk from Egocentric Robot Images?" *Workshop on Embodied Multimodal Reasoning (EMR)* at the *European Conference on Computer Vision (ECCV)*, 2026 (peer-reviewed, non-archival).
   arXiv:2608.12515
2. Spitkovska, V., Kuzmenko, D. "Trajectory Design and Budgeted Querying for Digital Twin Calibration." *Workshop on Curated Data for Efficient Learning (CDEL)* at the *European Conference on Computer Vision (ECCV)*, 2026 (peer-reviewed, non-archival).
   arXiv:2608.08631

Preprints

1. Kuzmenko, D., Shvai, N. "TD-MPC-Opt: Distilling Model-Based Multi-Task Reinforcement Learning Agents." arXiv:2507.01823, 2025 (under review).
   arxiv.org/abs/2507.01823 | code

Work in Progress

1. *This/That: Spatial Reference Resolution for Vision-Language Models* (AIRlab, University of Turin; contributor, ongoing) — how well VLMs and 3D pipelines resolve deictic reference ("this"/"that") from a person's egocentric viewpoint.

Workshop Papers & Posters

1. Kuzmenko, D., et al. "Semi-Supervised Learning for Efficient Visual Perception of Human-Robot Walking Environments." *Toronto Robotics Conference (TRC)*, 2024; *International Conference on Advanced Intelligent Robotics (ICAIR)*, 2023; *ICRA Workshop on Computer Vision in Robotics Research (CVWR)*, London, UK, 2023.
   researchgate.net | ICAIR pdf | CVWR pdf

Supervision & Student Research Program

Supervisor of **1 MSc thesis, 12 BSc theses, and 10 course papers** at NaUKMA, 2024 – 2026, spanning reinforcement learning, embodied perception, multi-agent systems, and vision-language models. Two supervised projects have produced peer-reviewed workshop papers at ECCV 2026.

Selected Supervised Research

- **O. Severhin** — *Efficient Policy Learning via Knowledge Distillation for Robotic Manipulation.* MSc thesis, 2025; co-supervised with Dr. Nadiya Shvai. Compute-efficient policy transfer for manipulation, extending the distillation line of my own model-based RL work.
- **V. Spitkovska** — *Active Reinforcement Learning for Data-Efficient Digital Twin Calibration.* BSc thesis, 2026. Active trajectory selection under a fixed query budget for sim-to-real parameter identification; published at the CDEL workshop, ECCV 2026.
- **V. Rudas** — *Vision-Language Models for Proxemics in Robot Perception.* BSc thesis, 2026. Whether VLMs can judge socially and physically unsafe proximity from a robot's own camera view; published at the EMR workshop, ECCV 2026.
- **T. Matiichyk** — *Ego-Centric Vision for Hand Occupancy Detection: Toward Safer Human-Robot Interaction.* BSc thesis, 2026. Egocentric perception of whether a person's hands are free, as a precondition for safe handover and physical interaction.
- **V. Kryvych** — *Assessing Robustness of Vision-Language Models to Domain and Distribution Shifts.* Course paper, 2026. Systematic evaluation of VLM reliability under domain shift.
- **Y. Vinokur** — *Cooperation of Partially Observed Agents in Ad-Hoc Open Teams.* BSc thesis, 2025. Benchmarked eight decentralised multi-agent RL paradigms under partial observability and open-team composition; ported the Wildfire benchmark to PettingZoo.
- **V. Beimuk** — *Energy Conservation in Autonomous Agents Using Reinforcement Learning.* BSc thesis, 2025; co-authored journal publication. Soft Actor-Critic with fuel-penalised reward shaping, studying speed/energy trade-offs in learned driving policies.
- **V. Bezborodov** — *Object Detection Model Pruning with Application to Human Recognition in UAV Footage.* BSc thesis, 2025. Quantisation, structured pruning, and distillation for real-time on-board inference.

Robotic Platforms, Systems & Technical Skills

- **Physical platforms.** *Reachy Mini* – deployed and integrated the platform as a shared experimental robot at AIRlab, UniTo, where it now serves as hardware and inspiration for a Python course for non-technical master's students, and for a supervised thesis student teaching English to children. *Museum tour-guide robot* – partook in the 3D scene-mapping pipeline for autonomous museum navigation. *Robotic exoskeletons and wearable locomotion systems* – perception pipelines developed with the Machine Intelligence Lab.
- **Open-source research software.** Released and maintain EED-Gym benchmark and evaluation suite and TD-MPC-Opt distillation framework, both with reproduction code.

Research methods: model-based RL, world models, policy distillation and quantisation, imitation learning, mixture-of-experts policies, VLA/VLM systems, embodied-AI benchmarking, HRI evaluation

Simulation & robotics tooling: MuJoCo, Gymnasium, Isaac Gym/Lab, Genesis, PyBullet, Habitat, PettingZoo

Programming & ML: Python, PyTorch, HuggingFace, TensorFlow, Docker, Git, distributed training

Systems & deployment: vLLM, low-latency inference, agentic pipelines, cloud deployment (Azure / AWS / GCP), production ML

Invited Talks & Conference Presentations

When Robots Say No: The Empathic Ethical Disobedience Benchmark Apr 2026
Presentation at *COMETE Workshop*, University of Turin, Turin, Italy

When Robots Say No: The Empathic Ethical Disobedience Benchmark Mar 2026
Main-track paper presentation at *International Conference on Human-Robot Interaction (HRI)*, Edinburgh, Scotland

Robotics and AI research Sep 2025
Talk and panel participation at *Kyiv AI & BigData Day 2025*, Kyiv, Ukraine

Knowledge Transfer in Model-Based RL Agents Jun 2025
Invited talk, RL Reading Club at *University of Southampton*, Southampton, UK

Knowledge Transfer in Model-Based RL Agents for Efficient Multi-Task Learning May 2025

Poster at *24th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, Detroit, MI, USA

SSL for Efficient Perception of Human-Robot Walking Environments 2023 – 2024

Presented at *Toronto Robotics Conference (TRC)*, *ICRA Workshop on Computer Vision in Robotics Research (CVWR)*, and *International Conference on Advanced Intelligent Robotics (ICAIR)*

Academic Service

- Reviewer, *Workshop on Curated Data for Efficient Learning at ECCV 2026*
- Reviewer, *ICTERI 2026*
- Reviewer, *International Conference on Human-Robot Interaction 2026*
- Reviewer, *International Conference on Social Robotics 2026*
- Reviewer, *Behaviour & Information Technology*
- Reviewer, *Journal of Field Robotics*

Teaching Experience

Reinforcement Learning *MSc (Graduate)*
2023 – 2025 3 offerings

Machine Learning *MSc (Graduate)*
2022 – 2026 4 offerings

Computer Vision *MSc (Graduate)*
2022 – 2025 4 offerings

Image Analysis & Computer Vision *MSc (Graduate)*
2024 – 2026 3 offerings

NLP Systems *MSc (Graduate)*
2024 – 2025 2 offerings

Deep Learning for Computer Vision *BSc (Undergraduate)*
2024 – 2026 3 offerings

Machine Learning *BSc (Undergraduate)*
2024 – 2025 2 offerings

Advanced Machine Learning *MSc (Graduate)*
2024 1 offering *SET University, Kyiv*

Total: 8 distinct courses across 22 offerings (October 2022 – present).

Industry & Engineering Experience

Nine years of software and applied-AI engineering, conducted alongside research.

Head of AI, OnlyMonster (Remote) *Jan 2025 – present*

Founding AI engineer; established the AI function and set its technical direction. Production language and agentic systems for real-time assistance, multilingual translation, and content safety, including low-latency inference and moderation pipelines at scale.

Data Science Engineering Manager, Litslink (Kyiv, Ukraine) *Nov 2022 – Dec 2024*

Led applied ML projects across computer vision, predictive modelling, and LLM pipelines (over 25 production and PoC projects, including Sportsbox AI).

Machine Learning Engineer, Infopulse (Kyiv, Ukraine) *Nov 2021 – Nov 2022*

Automated segmentation and OCR for presentation slides; built a technical-diagram processing pipeline (mmdet, OpenCV) with BFS-based graph reconstruction; optimised inference by ∼70%.

Outreach & Professional Service

- Technical Committee Member, AI HOUSE Oct 2024 – Sep 2025
- Delivered over 20 talks at AI/ML meetups and conferences in Ukraine and abroad
- Machine learning mentoring: Projector Institute, Women Who Code Kyiv 2022 – 2023

Languages

Ukrainian: Native, *English:* C2, *German:* B2, *Italian:* B1

References

Available upon request.